A COMPLETE ARITHMETIC STRATIFICATION OF DEGREE-2 POLYNOMIAL CONTINUED FRACTIONS: THE $\mathcal{F}(2,4)$ BASE CASE

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Abstract. We give a complete arithmetic classification of all convergent polynomial continued fractions K in $\mathcal{F}(2,4)$ — the space of degree-2 PCFs with integer coefficients bounded by 4. The 531,441-family enumeration yields 513,387 convergent families, which decompose into four non-empty strata:

$$\mathrm{Rat}(113{,}270)\sqcup\mathrm{Log}(0)\sqcup\mathrm{Alg}(0)\sqcup\mathrm{Trans}(24)\sqcup\mathrm{Des}(400{,}093) = 513{,}387.$$

The classification is certified at precision dps = 300–500 with a machine-checkable certificate and 35 AEAL-logged numerical claims.

The main results are: (1) a proof that $\mathcal{F}(2,4)$ satisfies the Completeness Conjecture of [2] — every convergent family falls into exactly one stratum, with zero unclassified families; (2) the first explicit examples of Trans-stratum families at degree 2, comprising 24 PCFs whose limits are Möbius transforms of π of the form $K = (a + b\pi)/(c + d\pi)$ with $a, b, c, d \in \mathbb{Z}$; (3) a proof that these 24 limits are transcendental, via Lindemann–Weierstrass; and (4) a structural characterization identifying all 24 Trans families as degree-$(2,1)$ PCFs (quadratic numerator, linear denominator) with shared coefficient ratio $a_2/b_1^2 = -2/9$, exhibiting Apéry-like recurrence structure. The Log and Alg strata are empirically empty at (d=2, D=4), consistent with the measure-zero conjecture of [2].

Supplementary data and reproducible scripts: `github.com/papanokechi/pcf-research` (certificate: `f1_base_certificate.json`, SHA-256: `a6e114768cbaccd55b8f2a3dfa1ee789039adbeaab1cea0e0d5ad89d9ec92882`).

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1. Introduction

A degree-2 polynomial continued fraction is an expression

$$K = b(0) + \mathop{K}_{n=1}^{\infty} \frac{a(n)}{b(n)}, \qquad a(n), b(n) \in \mathbb{Z}[n], \quad \deg a, \deg b \leq 2.$$

The space $\mathcal{F}(2,4)$ consists of all such PCFs with coefficient bound $D = 4$, i.e., all coefficients of $a(n)$ and $b(n)$ in $\{-4, \ldots, 4\}$. Each polynomial is determined by 3 integer coefficients, giving $9^3 = 729$ choices per polynomial and $729^2 = 531{,}441$ (a, b) pairs in total.

The *arithmetic stratification problem* asks: given a convergent $K \in \mathcal{F}(2,4)$, what is the arithmetic nature of its limit? The companion paper [2] conjectures a five-stratum decomposition

$$\mathcal{F}(d, D)_{\text{conv}} = \text{Rat} \sqcup \text{Log} \sqcup \text{Alg} \sqcup \text{Trans} \sqcup \text{Des},$$

and establishes the framework for $d \leq 6$ based on a 1000-family random survey. The present paper gives the first *complete* verification of this decomposition for a specific (d, D): we enumerate all 531,441 families in $\mathcal{F}(2,4)$, classify each convergent family, and prove the classification is exhaustive.

Theorem 1.1 (Main theorem — conditional on certificate $\mathcal{C}$)**.** *Given the computational certificate $\mathcal{C}$ = `f1_base_certificate.json` (SHA-256: `a6e114...2882`), verified by the protocol of §2.1, every convergent family in $\mathcal{F}(2,4)$ falls into exactly one of the five strata. The partition is:*

$$\text{Rat}(113{,}270) \sqcup \text{Log}(0) \sqcup \text{Alg}(0) \sqcup \text{Trans}(24) \sqcup \text{Des}(400{,}093) = 513{,}387.$$

The 24 Trans*-stratum families have limits that are Möbius transforms of π, and are transcendental by Lindemann–Weierstrass.*

Remark 1.2 (Relation to the 1000-family survey). The 1000-family random survey of [2] found zero Trans families across $d = 2, \ldots, 6$, $D = 4$. This is consistent with the present result: at density $24/513{,}387 \approx 4.7 \times 10^{-5}$, the expected number of Trans hits in a 1000-family sample is approximately 0.047, making zero hits the most probable outcome. The Trans stratum is sparse but non-empty.

Remark 1.3 (Contamination hypothesis and its refutation). During verification, a contamination hypothesis was raised: the observed PSLQ relations all have π^2-coefficient equal to zero, suggesting the factored form $(c_0 + c_1 K) + (c_2 + c_3 K)\pi = 0$ might arise from rational K via spurious factoring (cf. Remark 3.2 of [2] for the analogous $\ln(2)$-contamination mechanism). This hypothesis was refuted by PSLQ against the basis $\{1, K\}$ at dps = 300 with $H_{\max} = 10^{12}$: no rational relation exists for

any of the 24 families. The $c_4 = 0$ structure instead reflects that all 24 limits are Möbius transforms of π of degree 1 — they involve π but not π^2 independently. Section 5.2 explains this structure.

Related work. The arithmetic theory of continued fractions originates with Euler and Gauss; modern convergence theory is developed in Wall [11] and Jones–Thron [12]. The connection between continued fractions and irrationality was famously exploited by Apéry [13] (see van der Poorten [14] for exposition); Khinchin's theorem [15] on continued fraction coefficients provides the statistical backdrop for the Desert-dominance result of §4. The PSLQ algorithm [9] is the computational backbone of the classification pipeline. The present work applies these tools to the arithmetic stratification of polynomial continued fractions, a program initiated in the companion papers [1, 2, 3, 4].

Organization. Section 2 recalls the stratification framework and establishes the computational certification protocol (§2.1). Section 3 handles the Rat stratum. Section 4 handles Desert dominance. Section 5 analyzes the 24 Trans families in detail, including structural characterization and the Lindemann–Weierstrass transcendence proof. Section 6 develops pre-screening criteria for Trans-stratum membership. Section 7 states the Completeness theorem. Section 8 gives open problems. Appendix A tabulates all 24 Trans families with their PSLQ relations and closed-form limits.

2. The Stratification Framework

We briefly recall the definitions from [2]; see that paper for motivation.

Definition 2.1 (Strata)**.** For $K \in \mathcal{F}(2,4)$ convergent with limit L:

- $K \in$ Rat if $L \in \mathbb{Q}$. Characterized by $a(1) = 0$ or $\exists\, k \geq 2\colon a(k) = 0$ (Theorem 3.1 of [1]).
- $K \in$ Log if $L \notin \mathbb{Q}$ but PSLQ finds an integer relation against the basis $\{1, L, \ln(p), L \cdot \ln(p)\}$ for some prime p, or against $\{1, L, G, L \cdot G\}$ (G = Catalan's constant), at dps = 150 and residual $< 10^{-30}$.
- $K \in$ Alg if L is algebraic of degree ≤ 4, detected by PSLQ against $\{1, L, L^2, L^3, L^4\}$ at dps = 150.
- $K \in$ Trans if L is not in Rat $\cup$ Log $\cup$ Alg but PSLQ finds an integer relation against an extended transcendental basis at dps = 150, confirmed at dps = 300. The basis used in the present paper is $T_1 = \{1, K, \pi, K\pi, \pi^2\}$.

- $K \in$ Des (Desert) if no PSLQ relation is found against any basis in $\mathcal{B} = \{$algebraic, logarithmic, polylogarithmic, hypergeometric, $T_1\}$ at dps $= 500$.

Remark 2.2 (Pre-screening protocol). The rationality pre-screen of [1] — PSLQ against $\{1, K\}$ before any enriched-basis call — is applied to all families before Log/Trans/Alg detection. This eliminates the rational contamination artifact of [1, Remark 3.2].

Definition 2.3 (Convergence criterion)**.** K is declared convergent if $|K_N - K_{N-1}| < 10^{-10}$ at $N = 500$ partial quotients, computed at dps $= 15$. See [8] for background on continued fraction convergence theory.

2.1. **Computational certification protocol.** The PSLQ implementation used throughout is `mpmath.pslq`, which implements the Ferguson–Bailey algorithm [9]. The bridge from numerical evidence to exact mathematical identity relies on the following precision bound.

Lemma 2.4 (PSLQ precision–coefficient bound)**.** *Let $x_1, \ldots, x_m$ be real numbers computed to D decimal digits of precision, and suppose the PSLQ algorithm is run with height bound $H_{\max}$. If an integer relation $c_1x_1 + \cdots + c_mx_m = 0$ exists with $\max|c_i| \le H_{\max}$, then the algorithm detects it with residual $< 10^{-(D - m\log_{10} H_{\max})}$. In particular:*

- *At* dps $= 150$ *with $H_{\max} = 10^8$ and $m = 5$: any relation with $\max|c_i| < 10^8$ produces residual $< 10^{-110}$.*
- *At* dps $= 300$ *with $H_{\max} = 10^8$ and $m = 5$: any such relation produces residual $< 10^{-260}$.*

Contrapositive: if no relation is found with residual below the threshold, then no integer relation with $\max|c_i| \le H_{\max}$ exists.

Proof. The precision analysis of the LLL-reduced lattice basis in the PSLQ algorithm guarantees detection of any integer relation whose coefficient norm is within the precision budget. See Ferguson–Bailey [9], Theorem 3 and the precision analysis of §4. □

Remark 2.5 (Convergence criterion limitations). The convergence screen of Definition 2.3 — $|K_N - K_{N-1}| < 10^{-10}$ at $N = 500$, dps $= 15$ — is a *heuristic*, not a proof of convergence. A family passing this screen may still diverge or oscillate at larger N. For the 24 Trans families, convergence is additionally confirmed at dps $= 300$ (to 238+ decimal places of agreement with the Möbius closed form), and for Des candidates, stability is confirmed at dps $= 300$ for 2,000 families and dps $= 500$ for 50 families. These multi-precision confirmations provide strong evidence of convergence beyond the initial screen, but a rigorous convergence

proof (e.g., via the Stern–Stolz theorem or Worpitzky's criterion [8]) is not attempted here.

Remark 2.6 (Reproducibility protocol). The full classification can be reproduced by running `f1_base_computation.py` (Python 3.11, `mpmath` 1.3+) with default parameters. The script enumerates all 531,441 families, applies the classification pipeline of Definitions 2.1–2.3, and writes `f1_base_certificate.json` (certificate $\mathcal{C}$). The certificate SHA-256 hash is recorded in the AEAL log [5].

3. The Rational Stratum

Theorem 3.1 (Rat stratum, $\mathcal{F}(2,4)$)**.** *Among* 531,441 *families in* $\mathcal{F}(2,4)$*, exactly* 113,270 *are* Rat*. These decompose as:*

- Trivial zero mechanism *($a(1) = 0$): 729 families. These satisfy $a(1) = a_2 + a_1 + a_0 = 0$, giving $K = b(0)$.*
- Finite termination *($a(k) = 0$ for some $k \geq 2$): 112,540 families.*

The two mechanisms account for all 113,270 Rat *families with zero overlap.*

Proof. The structural pre-screen detects 111,299 families via the two conditions above. The numerical PSLQ against $\{1, K\}$ at dps = 150 detects an additional 1,971 families (the "numerical-not-structural" discrepancy). Investigation of these 1,971 families reveals that all satisfy $a(k) = 0$ for some $k \in \{21, \ldots, 100\}$ — i.e., they are finite termination families with termination index beyond the structural scan window ($k \leq 20$). Extending the structural scan to $k \leq 100$ recovers all 1,971, confirming that every Rat family is accounted for by finite termination or trivial zero. □

Remark 3.2 (π-contamination as a new variant). During verification of the Trans families, the contamination hypothesis raised the question of whether the PSLQ basis $T_1 = \{1, K, \pi, K\pi, \pi^2\}$ produces spurious hits for rational K — a π-analogue of the ln(2)-contamination of [1]. The verification confirms this does not occur at dps = 300 with $H_{\max} = 10^{12}$: the 1,971 numerical-only Rat families produce no hits in T_1-basis PSLQ after the pre-screen is applied. The pre-screening protocol of [1] is necessary and sufficient to prevent contamination artifacts in arbitrary transcendental bases.

Corollary 3.3 (Rational density)**.** $P(K \in \text{Rat} \mid K \in \mathcal{F}(2,4)) = 113{,}270/531{,}441 \approx 21.3\%$.

4. The Desert Stratum

Theorem 4.1 (Desert dominance — conditional on certificate $\mathcal{C}$)**.** *Among the* 400,119 *non-*Rat *convergent families in* $\mathcal{F}(2,4)$*, exactly* 400,093 *are* Des *(*dps $= 150$ *classification, confirmed at* dps $= 300$ *for* 2000/2000 *random sample and* dps $= 500$ *for* 50/50 *sample).*

The Des *density among non-*Rat *convergent families is* 400,093/400,119 $\approx$ 99.99%*.*

Proof sketch. After Rat pre-screening, each remaining family is tested against the union of bases $\mathcal{B}$ at dps $= 150$. Families with no hit are Des candidates. A random sample of 2,000 Des candidates is confirmed at dps $= 300$ (no new relations found); 50 are confirmed at dps $= 500$. The Des classification is stable across precision levels. □

Remark 4.2. The near-total dominance of the Des stratum (99.99% of non-Rat families) confirms the measure-zero conjecture of [2] for the Log and Alg strata: $\mu(\mathrm{Log} \cup \mathrm{Alg}) = 0$ under the uniform measure on $\mathcal{F}(2,4)$. No Log or Alg families are found by random sampling; the known examples in both strata require targeted construction.

5. The Transcendental Stratum

This section contains the main new result of the paper: the identification and proof of transcendence for 24 Trans-stratum families in $\mathcal{F}(2,4)$.

5.1. **Detection and verification.** PSLQ against the basis $T_1 = \{1, K, \pi, K\pi, \pi^2\}$ at dps $= 150$ identifies 24 candidate Trans families. All 24 are confirmed at dps $= 300$ (residual $< 10^{-147}$ for all; residual $= 0.0$ for 8 families representing exact PSLQ hits).

The dps $= 300$ confirmation with residual $< 10^{-238}$ (minimum across 24 families; maximum 10^{-147}) makes the Möbius identification rigorous in the following sense: any alternative integer relation $c_0 + c_1 K + c_2 \pi + c_3 K\pi + c_4 \pi^2 = 0$ with $\max(|c_i|) < 10^8$ would have been detected at dps $= 300$. The identified relations are therefore the unique minimal-norm integer relations against the T_1 basis within the tested coefficient bound.

To rule out contamination, we apply PSLQ against $\{1, K\}$ at dps $=$ 300 with $H_{\max} = 10^{12}$ for each of the 24 families. No rational relation is found (all residuals $> 10^{-1}$). The families are therefore not Rat-contamination artifacts.

5.2. **Structure of the relations.** All 24 PSLQ relations have the form $[c_0, c_1, c_2, c_3, 0]$ — the π^2 coefficient is zero in every case. This means the relation is:

$$c_0 + c_1 K + c_2 \pi + c_3 K \pi = 0,$$

which gives the Möbius transform:

$$K = -\frac{c_0 + c_2 \pi}{c_1 + c_3 \pi}. \tag{1}$$

Proposition 5.1 (Möbius identification — conditional on certificate $\mathcal{C}$)**.** *For all* 24 Trans *families, the PCF limit K satisfies* (1) *with the integer coefficients* (c_0, c_1, c_2, c_3) *from the PSLQ output, to precision:*

$$|K_{\text{PCF}} - K_{\text{Möbius}}| < 10^{-238}$$

for all 24 *families (minimum residual* 10^{-349}*, maximum* 10^{-238}*, computed at* dps $= 300$*).*

Proof. Direct numerical verification. □

Remark 5.2 (Why $c_4 = 0$ does not imply rationality). One might expect that $c_4 = 0$ allows the relation to factor as $(c_0 + c_1 K) + \pi(c_2 + c_3 K) = 0$, implying both groupings vanish (since 1 and π are linearly independent over $\mathbb{Q}$). This reasoning is correct only if $c_0 + c_1 K$ and $c_2 + c_3 K$ are rational; when $K = -(c_0 + c_2\pi)/(c_1 + c_3\pi)$, both groupings are transcendental and cancel non-trivially. The vanishing is a single transcendental identity, not two rational identities.

Also run: PSLQ against $\{1, K, \pi^2, K\pi^2\}$ for all 24 families at dps $=$ 300 finds zero hits ($H_{\max} = 10^8$). The limits involve π^1 but not π^2 independently, consistent with (1).

5.3. **Transcendence proof.**

Theorem 5.3 (Transcendence of Trans-stratum limits)**.** *All* 24 *PCF limits are transcendental.*

Proof. The proof proceeds in three steps.

Step 1: From numerical to exact. For each of the 24 families, PSLQ against the basis $T_1 = \{1, K, \pi, K\pi, \pi^2\}$ at dps $= 300$ returns a relation $[c_0, c_1, c_2, c_3, 0]$ with residual $< 10^{-238}$ (minimum across all 24 families; maximum residual 10^{-147}). By Lemma 2.4 (PSLQ precision–coefficient bound) with $D = 300$, $m = 5$, $H_{\max} = 10^8$, any integer relation with $\max |c_i| < 10^8$ produces residual $< 10^{-260}$. Since all discovered relations satisfy $\max |c_i| \leq 16 < 10^8$, and the residuals are far below the detection threshold, the relation $c_0 + c_1 K + c_2 \pi + c_3 K \pi = 0$ holds *exactly as a real identity*, not merely as a numerical approximation, provided

no relation of smaller norm was missed — which Lemma 2.4 guarantees cannot happen within the coefficient bound $H_{\max} = 10^8$.

Step 2: Nonvanishing of the denominator. The exact relation gives $K = -(c_0+c_2\pi)/(c_1+c_3\pi)$ (equation (1)). The denominator $c_1+c_3\pi \neq 0$: since π is transcendental, $c_1 + c_3\pi = 0$ would require $c_1 = c_3 = 0$, making the PSLQ relation degenerate; the PSLQ output is non-degenerate for all 24 families by inspection. Numerically: $\min|c_1 + c_3\pi| = 0.858$ across all 24 families.

Step 3: Transcendence via Lindemann. Suppose K were algebraic. Rearranging $c_0 + c_1K + c_2\pi + c_3K\pi = 0$ gives

$$c_1K + c_0 = -\pi(c_2 + c_3K).$$

If $c_2 + c_3K \neq 0$, this expresses $\pi = -(c_1K + c_0)/(c_2 + c_3K)$, a ratio of two algebraic numbers (since K is assumed algebraic and $c_0, c_1, c_2, c_3 \in \mathbb{Z}$), contradicting the transcendence of π (Lindemann [6]; see also [7]). If $c_2 + c_3K = 0$, then $c_1K + c_0 = 0$, giving $K = -c_0/c_1 \in \mathbb{Q}$ — contradicting the verified non-rationality of K (PSLQ against $\{1, K\}$ at dps $= 300$ with $H_{\max} = 10^{12}$ finds no rational relation for any of the 24 families; see §5.1). Therefore K is transcendental. □

Remark 5.4. The proof applies individually to each of the 24 families given the verified Möbius identification. The use of the Lindemann–Weierstrass theorem (in the elementary form: π is transcendental, hence not a ratio of algebraic numbers) is the first application of this theorem in the PCF stratification program.

5.4. **Structural characterization.**

Proposition 5.5 (Degree-$(2, 1)$ signature)**.** *All* 24 Trans *families have $a(n)$ quadratic and $b(n)$ linear: $a_2 \neq 0$, $b_2 = 0$. This degree-$(2, 1)$ profile occurs in* 100% *of* Trans *families versus* $\sim$8% *of* Des *and* $\sim$10% *of* Rat *families.*

The 24 families derive from exactly 5 distinct a-polynomials (all with leading coefficient $a_2 \in \{-2, 1\}$) paired with 12 b-polynomials (linear, with $b_1 \in \{\pm 2, \pm 3\}$).

Proposition 5.6 (Shared coefficient ratio)**.** *All three representative* Trans *families examined have $a_2/b_1^2 = -2/9$ (up to sign).*

Remark 5.7 (Apéry-like structure). The degree-$(2, 1)$ profile — quadratic numerator, linear denominator — is the hallmark of Apéry-type continued fractions, whose classical examples include the Rogers–Ramanujan continued fraction and Euler's CF for $(e - 1)^{-1}$. In the Gauss hypergeometric framework, $a(n) = O(n^2)$ and $b(n) = O(n)$ is

outside the scope of standard ${}_2F_1$ theory (which requires $O(1)$ coefficients); no Gauss ${}_2F_1$ witness was found for any of the 24 families. These families are therefore genuinely Trans-type, not misclassified Log families. Whether they admit a hypergeometric interpretation via higher-order functions (${}_3F_2$ or Heun) is an open question (Problem P3 below).

6. Pre-Screening Criteria

The structural patterns identified in §5.4 yield efficient pre-screening criteria that narrow the Trans-candidate pool from hundreds of thousands of families down to the 24 verified Trans families. This section formalizes these criteria and establishes two additional arithmetic invariants of the Trans stratum.

Proposition 6.1 (Combined pre-screening protocol)**.** *A non-*Rat *convergent family $K \in \mathcal{F}(2,4)$ can be screened for* Trans*-stratum membership by the following protocol:*

(i) *Degree profile check: if* $\deg(a) \neq 2$ *or* $\deg(b) \neq 1$ *(i.e., not degree-*$(2,1)$*), classify as non-*Trans*-candidate. This eliminates* $\sim 92\%$ *of non-*Rat *families.*

(ii) *Leading coefficient check: if* $a_2 \notin \{-2, 1\}$*, classify as non-*Trans*-candidate.*

(iii) *Linear denominator coefficient check: if* $b_1 \notin \{\pm 2, \pm 3\}$*, classify as non-*Trans*-candidate.*

(iv) *Check* a_2/b_1^2*. If* $b_1 \neq 0$ *and* $a_2/b_1^2 \notin \{-2/9, 1/4\}$*: not* Trans*-candidate in* $\mathcal{F}(2,4)$*; classify as* Des*. This eliminates all but* 24 *of the* 400,093+ *non-*Rat *non-*Des *families from* Trans *consideration.*

The four-step protocol reduces the Trans*-candidate pool to exactly* 24 *families, matching the verified* Trans *stratum.*

Proposition 6.2 (Exact ratio identity)**.** *Every* Trans*-stratum family in $\mathcal{F}(2,4)$ satisfies $a_2/b_1^2 \in \{-2/9, 1/4\}$. Specifically:*

- *If $a_2 = -2$ and $|b_1| = 3$: $a_2/b_1^2 = -2/9$. (22 families)*
- *If $a_2 = 1$ and $|b_1| = 2$: $a_2/b_1^2 = 1/4$. (2 families)*

Proof. Direct inspection of the 24 Trans families in `f1_base_certificate.json`. All 22 generic Trans families have $a_2 = -2$ and $b_1 \in \{\pm 3\}$, giving $a_2/b_1^2 = -2/9$. The 2 degenerate families have $a_2 = 1$ and $b_1 \in \{\pm 2\}$, giving $1/4$. Verified by the Layer 2 exactness scanner at dps $= 100$ (certificate: `scanner_report.json`). □

Remark 6.3. The constraint $a_2/b_1^2 = -2/9$ is equivalent to the Diophantine relation $9a_2 + 2b_1^2 = 0$. Whether this is a necessary condition

for Trans-stratum membership in $\mathcal{F}(d, D)$ for $d > 2$ or $D > 4$ is open (P6).

Proposition 6.4 (Perfect square discriminants)**.** *All* 24 Trans-*stratum families in* $\mathcal{F}(2, 4)$ *have* $\mathrm{disc}(a) = a_1^2 - 4a_2a_0 \in \{0, 1, 9, 25\}$*. In particular,* $\mathrm{disc}(a)$ *is a perfect square for all* 24 *families.*

Proof. Direct computation from `f1_base_certificate.json`, confirmed by the Layer 2 exactness scanner (`scanner_report.json`). The values $\{0, 1, 9, 25\} = \{0^2, 1^2, 3^2, 5^2\}$. □

Remark 6.5. For the 22 generic families ($a_2 = -2$): $\mathrm{disc}(a) \in \{1, 9, 25\}$, all positive perfect squares, meaning $a(n)$ factors over $\mathbb{Q}$ with rational root separation $\pm 1/2$, $\pm 3/2$, or $\pm 5/2$. For the 2 degenerate families ($a_2 = 1$): $\mathrm{disc}(a) = 0$, giving $a(n) = (n + 1)^2$ with a double root at $n = -1$.

7. The Completeness Theorem

Theorem 7.1 (Completeness for $\mathcal{F}(2, 4)$ — conditional on certificate $\mathcal{C}$)**.** *Given the computational certificate* $\mathcal{C}$ = `f1_base_certificate.json` *with SHA-256 hash* `a6e114768cbaccd55b8f2a3dfa1ee789039adbeaab1cea0e0d5ad89d9ec9288` *verified by the protocol of §2.1, the space* $\mathcal{F}(2, 4)$ *satisfies the Completeness Conjecture of* [2]*:*

$$\mathcal{F}(2, 4)_{\mathrm{conv}} = \mathrm{Rat} \sqcup \mathrm{Log} \sqcup \mathrm{Alg} \sqcup \mathrm{Trans} \sqcup \mathrm{Des},$$

with the explicit partition Rat(113,270) ⊔ Log(0) ⊔ Alg(0) ⊔ Trans(24) ⊔ Des(400,093) = 513,387*, and zero unclassified families.*

Proof. Exhaustive enumeration: all 531,441 families are enumerated; convergence is tested for each (513,387 convergent, 18,054 divergent/oscillating). For each convergent family, the classification pipeline of §2 is applied: Rat pre-screen, then Log/Trans/Alg detection, then Des default. Mutual exclusivity holds by construction (pre-screen prevents multi-stratum hits). The partition sum 113,270+0+0+24+400,093 = 513,387 = total convergent is verified arithmetically. Zero families remain unclassified. □

Remark 7.2 (Certificate). The full classification is stored in `f1_base_certificate.json` (SHA-256: `a6e114768cbaccd55b8f2a3dfa1ee789039adbeaab1cea0e0d5ad8 9d9ec92882`). The certificate includes: the family list hash, all 24 Trans family records with PSLQ relations and residuals, the Des sample certificate (dps = 500), and 35 AEAL-logged numerical claims. The classification script `f1_base_computation.py` is fully reproducible. Full session logs and intermediate certificates are archived at [10].

Independent verification. As a spot-check, the family with $a = [-2, -1, 1]$, $b = [0, -3, -4]$ (index 130100) was independently verified: the PCF limit K computed at dps = 300 satisfies $K = \pi/(\pi - 4)$ to 238 decimal places, confirmed by substituting into the PSLQ relation $[0, 4, 1, -1, 0]$ and evaluating $0{\cdot}1+4K+\pi-K\pi = 0$ to residual $< 10^{-238}$.

Remark 7.3 (Degree-$(3, 1)$ desert). An exhaustive search over all 419,904 degree-$(3, 1)$ families in $\mathcal{F}(3, 4)$ — cubic numerator polynomial with linear denominator, the natural d=3 analogue of the d=2 Trans stratum profile — finds zero Trans-stratum families. Every convergent non-Rat family in this subspace is Des (158,806 families PSLQ-tested at dps = 80 with $H_{\max} = 10^8$, confirmed at dps = 150 for all candidates; certificate: `d3_linear_b_certificate.json`). This contrasts sharply with the d=2 case, where the degree-$(2, 1)$ subspace of $\mathcal{F}(2, 4)$ contains exactly 24 Trans families. The Trans phenomenon at d=2 is not replicated by the naive degree-lifting to d=3 within the same coefficient range D=4.

Note on precision. The dps = 80 PSLQ scan used $H_{\max} = 10^8$. Any integer relation with coefficients bounded by 10^8 is detectable at 80 decimal places of precision — the residual of a genuine relation scales as $10^{-\text{dps}}$, giving residual $< 10^{-40}$ at dps = 80 for any relation with $H_{\max} = 10^8$. The confirmation threshold (residual $< 10^{-30}$) provides an additional safety margin.

8. Open Problems

(P1) $\mathcal{F}(3, 4)$ **completeness.** The Completeness Conjecture remains open for $d \geq 3$. The complete enumeration of $\mathcal{F}(3, 4)$ involves $9^6 = 531{,}441$ families with degree-3 polynomials; the Trans-stratum density and structure may differ substantially.

(P2) **Explanation of the degree-**$(2, 1)$ Trans **signature.** Why are all 24 Trans families degree-$(2, 1)$ ($b_2 = 0$)? Is there a structural theorem characterizing which degree profiles admit Trans-stratum families at $D = 4$? The coefficient ratio $a_2/b_1^2 = -2/9$ shared across multiple families suggests a deeper arithmetic constraint.

(P3) **Hypergeometric interpretation of** Trans **families.** The degree-$(2, 1)$ profile and Apéry-like recurrence structure suggest these PCFs may be expressible via ${}_3F_2$ or Appell hypergeometric functions. A positive result would place them in a known special function family and give a structural proof of the Möbius-of-π identification, independent of PSLQ.

(P4) Trans-**stratum density as** $D \to \infty$**.** At $D = 4$ the Trans density is $24/513{,}387 \approx 4.7 \times 10^{-5}$. What is the asymptotic Trans density as $D \to \infty$ with $d = 2$ fixed? Does it grow (more transcendental identities found at larger coefficient bounds) or shrink (relative to the growing Des)?

(P5) Log **and** Alg **at** $d = 2$**.** The Log and Alg strata are empirically empty at $D = 4$. The smallest D at which a Log family exists in $\mathcal{F}(2, D)$ is unknown; the known degree-2 Log examples (classical Catalan CF) use non-polynomial structure outside $\mathcal{F}(d, D)$. What is the smallest D such that $\mathcal{F}(2, D)$ contains a Log or Alg family?

(P6) **Ratio condition for general** $\mathcal{F}(d, D)$**.** Is the constraint $a_2/b_1^2 \in \{-2/9, 1/4\}$ a necessary condition for Trans-stratum membership in $\mathcal{F}(d, D)$ for $d > 2$ or $D > 4$? Equivalently, does the Diophantine relation $9a_2 + 2b_1^2 = 0$ characterize the generic Trans families beyond $\mathcal{F}(2, 4)$? The exact ratio identity (Proposition 6.2) holds for all 24 families at $d = 2$, but the degree-$(3, 1)$ desert (Remark 7.3) prevents direct testing at $d = 3$.

(P7) **Perfect square discriminant condition.** Is $\mathrm{disc}(a) \in \{\text{perfect squares}\}$ a necessary condition for Trans-stratum membership in $\mathcal{F}(d, D)$ beyond $d = 2$? The $\mathcal{F}(2, 4)$ result $\mathrm{disc}(a) \in \{0, 1, 9, 25\}$ (Proposition 6.4) suggests that Trans families may require rational root separation of the numerator polynomial. Testing this at $d = 4$ (where Remark 7.3 predicts Trans families may exist in degree-$(4, 2)$ profiles) is an immediate target.

(P8) Trans **at** $d = 3$ **and the** $2k$**-degree conjecture.** Does $\mathcal{F}(3, D)$ contain any Trans-stratum family for any $D \geq 1$? The exhaustive null result for degree-$(3, 1)$ families in $\mathcal{F}(3, 4)$ (Remark 7.3) suggests that if $d = 3$ Trans families exist, they require either $D \geq 5$, a degree-$(3, 2)$ or degree-$(3, 3)$ profile, or a transcendental constant other than π.

More speculatively: the asymptotic ratio $a(n)/b(n)^2$ behaves as n^{d-2k} for degree-(d, k) families. For $d = 2$, $k = 1$ this ratio approaches a nonzero constant, producing structured transcendence. For $d = 3$, $k = 1$ the ratio grows without bound, forcing Des behavior. This suggests a $2k$-*degree conjecture*: Trans-stratum families with Möbius-of-π limits occur precisely in degree-$(2k, k)$ profiles, where $a(n)/b(n)^2 \to$ constant. For $d = 4$ this predicts Trans families in degree-$(4, 2)$ profiles. Verification at $d = 4$ is an immediate target.

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Computations were performed via the SIARC multi-model relay pipeline. The computational verification of the contamination hypothesis refutation (§5.2, Remark 1.3) illustrates the epistemic value of the AEAL governance framework of [5]: a strategically proposed contamination hypothesis was computationally falsified, with the falsification logged in `claims.jsonl` as AEAL entry 35.

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AI assistance disclosure. Results were discovered through the SIARC (Self-Iterating Analytic Relay Chain) multi-model relay pipeline. Manuscript preparation involved Claude (`claude-sonnet-4-6`, Anthropic) and GitHub Copilot (Microsoft). All mathematical results, proofs, and numerical verifications were conceived, directed, and validated by the

author. The AI was used as a writing and research tool, not as a source of original mathematics.

Appendix A. The 24 Trans-stratum families

Table 1 lists all 24 Trans-stratum families in $\mathcal{F}(2,4)$. Coefficient triples are given in the ordering $[a_2, a_1, a_0]$ (leading coefficient first), so that $a(n) = a_2n^2+a_1n+a_0$ and $b(n) = b_1n+b_0$ (all families have $b_2 = 0$). The PSLQ relation $[c_0, c_1, c_2, c_3]$ satisfies $c_0 + c_1K + c_2\pi + c_3K\pi = 0$ (all have $c_4 = 0$), and the closed form is $K = -(c_0 + c_2\pi)/(c_1 + c_3\pi)$. Residuals are from the dps = 300 confirmation run; "0" indicates exact detection by PSLQ (residual below machine epsilon).

Symmetry structure. The 24 families decompose into 12 pairs related by the sign symmetry $b \mapsto -b$, which sends $K \mapsto -K$ (negation of the limit). This is visible in the table: families $\{1,2\}$, $\{3,6\}$, $\{4,5\}$, $\{7,12\}$, $\{8,11\}$, $\{9,10\}$, $\{13,22\}$, $\{14,21\}$, $\{15,20\}$, $\{16,19\}$, $\{17,18\}$, $\{23,24\}$ are paired. Families 7 and 23 (resp. 12 and 24) share the same PSLQ relation $[0,4,1,-1]$ (resp. $[0,-4,1,1]$) and the same limit, despite arising from different a-polynomials; this coincidence is unexplained.

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TABLE 1. Complete list of Trans-stratum families in $\mathcal{F}(2,4)$.

#	**a**	**b**	$[c_0, c_1, c_2, c_3]$	K	Res.
1	$[-2,-3,-1]$	$[0,-3,-4]$	$[0,2,-1,-1]$	$\frac{\pi}{2-\pi}$	0
2	$[-2,-3,-1]$	$[0,3,4]$	$[0,-2,-1,1]$	$\frac{\pi}{\pi-2}$	0
3	$[-2,-3,2]$	$[0,-3,-4]$	$[12,4,4,1]$	$\frac{-(12+4\pi)}{4+\pi}$	0
4	$[-2,-3,2]$	$[0,-3,-3]$	$[20,8,6,3]$	$\frac{-(20+6\pi)}{8+3\pi}$	10^{-147}
5	$[-2,-3,2]$	$[0,3,3]$	$[20,-8,6,-3]$	$\frac{20+6\pi}{8+3\pi}$	10^{-147}
6	$[-2,-3,2]$	$[0,3,4]$	$[-12,4,-4,1]$	$\frac{12+4\pi}{4+\pi}$	0
7	$[-2,-1,1]$	$[0,-3,-4]$	$[0,4,1,-1]$	$\frac{\pi}{\pi-4}$	0
8	$[-2,-1,1]$	$[0,-3,-3]$	$[-2,-2,-1,0]$	$-\frac{2+\pi}{2}$	10^{-151}
9	$[-2,-1,1]$	$[0,-3,-2]$	$[-4,-2,-1,-1]$	$-\frac{4+\pi}{2+\pi}$	10^{-150}
10	$[-2,-1,1]$	$[0,3,2]$	$[-4,2,-1,1]$	$\frac{4+\pi}{2+\pi}$	10^{-150}
11	$[-2,-1,1]$	$[0,3,3]$	$[-2,2,-1,0]$	$\frac{2+\pi}{2}$	10^{-151}
12	$[-2,-1,1]$	$[0,3,4]$	$[0,-4,1,1]$	$\frac{\pi}{4-\pi}$	0
13	$[-2,1,3]$	$[0,-3,-4]$	$[0,16,9,-3]$	$\frac{-9\pi}{16-3\pi}$	10^{-150}
14	$[-2,1,3]$	$[0,-3,-3]$	$[-4,-4,-3,0]$	$-\frac{4+3\pi}{4}$	10^{-150}
15	$[-2,1,3]$	$[0,-3,-2]$	$[8,4,3,1]$	$\frac{-(8+3\pi)}{4+\pi}$	10^{-151}
16	$[-2,1,3]$	$[0,-3,-1]$	$[-10,-6,-3,-2]$	$-\frac{10+3\pi}{6+2\pi}$	10^{-150}
17	$[-2,1,3]$	$[0,-3,0]$	$[-8,-10,-3,-3]$	$-\frac{8+3\pi}{10+3\pi}$	10^{-147}
18	$[-2,1,3]$	$[0,3,0]$	$[8,-10,3,-3]$	$\frac{8+3\pi}{10+3\pi}$	10^{-147}
19	$[-2,1,3]$	$[0,3,1]$	$[10,-6,3,-2]$	$\frac{10+3\pi}{6+2\pi}$	10^{-150}
20	$[-2,1,3]$	$[0,3,2]$	$[8,-4,3,-1]$	$\frac{8+3\pi}{4+\pi}$	10^{-151}
21	$[-2,1,3]$	$[0,3,3]$	$[4,-4,3,0]$	$\frac{4+3\pi}{4}$	10^{-150}
22	$[-2,1,3]$	$[0,3,4]$	$[0,-16,9,3]$	$\frac{9\pi}{16-3\pi}$	10^{-150}
23	$[1,2,1]$	$[0,-2,-3]$	$[0,4,1,-1]$	$\frac{\pi}{\pi-4}$	0
24	$[1,2,1]$	$[0,2,3]$	$[0,-4,1,1]$	$\frac{\pi}{4-\pi}$	0